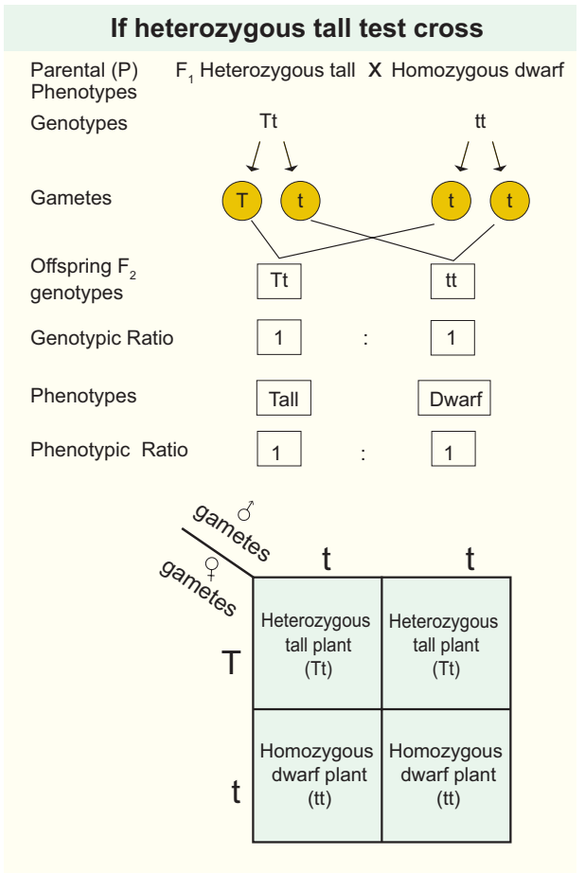


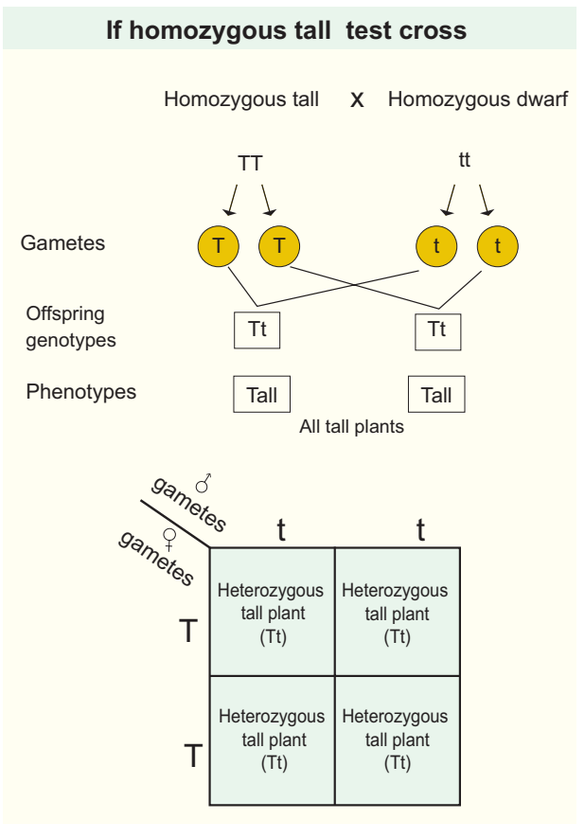


Figure 2.7: Test cross

Why Mendel's pea plants are tall and dwarf? Find out the molecular explanation.

Molecular characterization of Mendel's gene for plant height.

The plant height is controlled by a single gene with two alleles. The reason for this difference in plant height is due to the following facts: (i) the cells of the pea plant have the ability to convert a precursor molecule of gibberellins into an active form (GA1) (ii) Tall pea plants have one allele (Le) that codes for a protein (functional enzyme) which functions normally in the gibberellin-synthesis pathway and catalyzes the formation of gibberellins (GA1). The allele is dominant even if it is two (Le Le) or single (Le le), it produces gibberellins and the pea plants are tall. Dwarf pea plants have two recessive alleles (le le) which code for non-functional protein, hence they are dwarf.

Gene for plant height in Peas

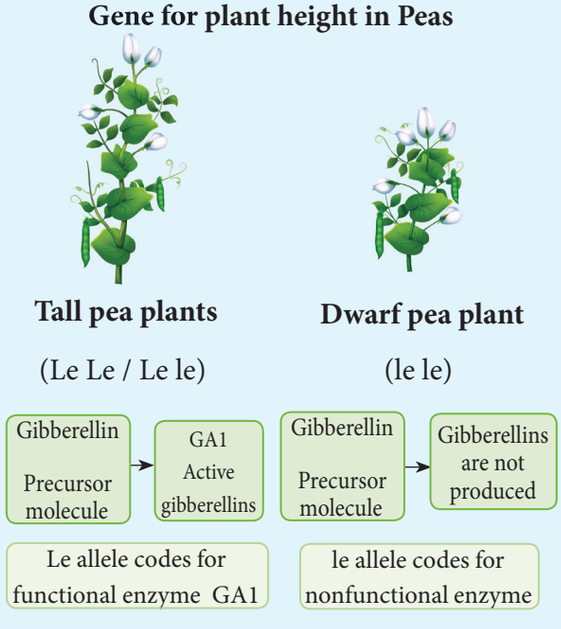


Figure 2.8: Gene for plant height in Peas

2.3.3 Back cross

- Back cross is a cross of F_1 hybrid with any one of the parental genotypes. The back cross is of two types; they are dominant back cross and recessive back cross.
- It involves the cross between the F_1 offspring with either of the two parents.

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